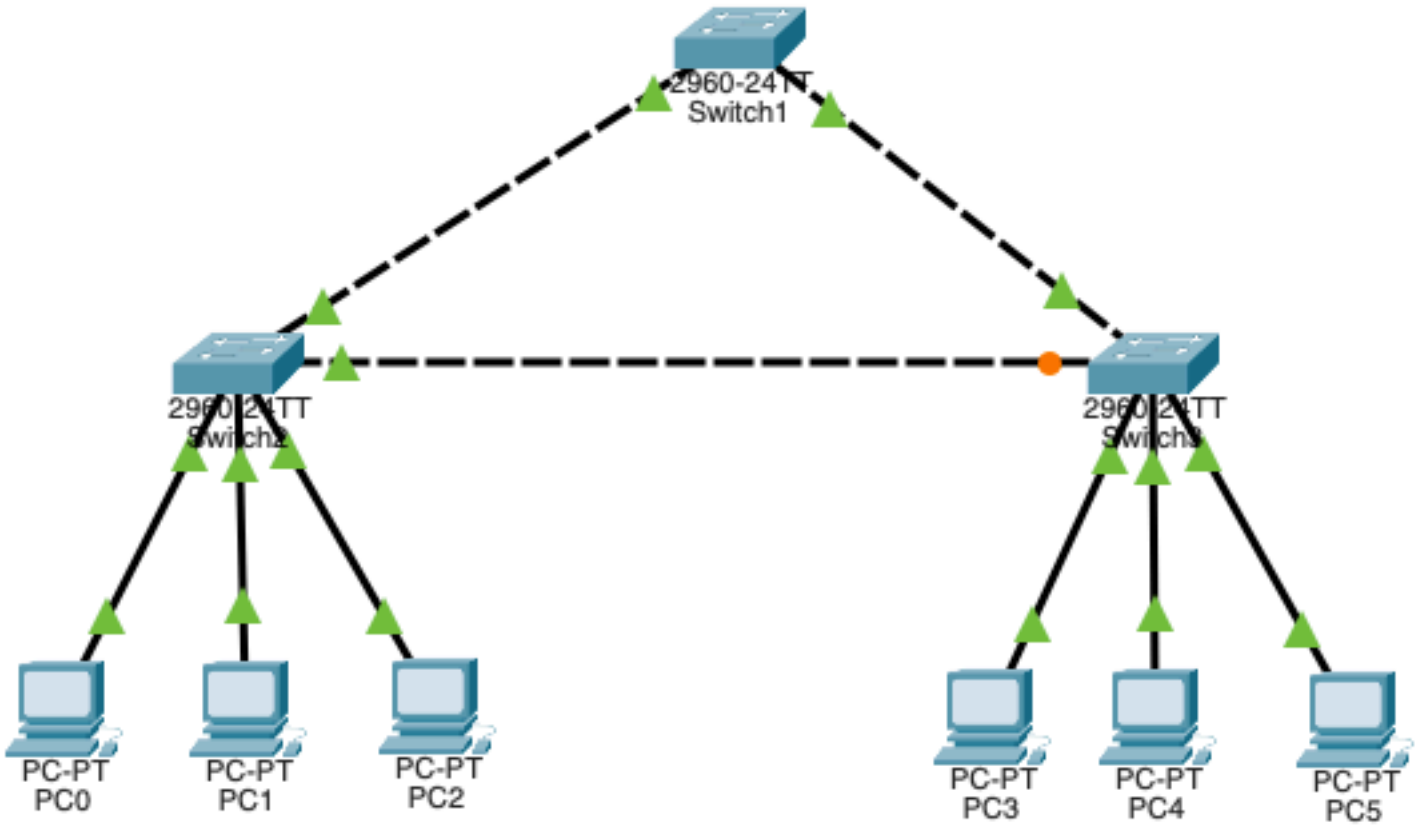


Built a simple network with three 2960 switches and six PCs to practice Spanning Tree and link failover.
Connected the switches in a triangle to provide a backup path if one link failed.
Assigned all PCs IP addresses in the same subnet so they could communicate without a router (192.168.10.0/24)
Enabled Rapid PVST on all switches to prevent switching loops and support fast recovery.
Configured switch priorities so S1 became the root bridge
Enabled PortFast on PC-facing ports so the PCs can communicate immediately.
Checked port roles with show spanning-tree vlan 1 to verify the root bridge and forwarding paths.
Confirmed S3's Fa0/4 was alternate/blocking to verify that Spanning Tree prevented a loop while retaining a backup path.
Pinged between the PCs to confirm connectivity before introducing a failure.
Shut down S3's Fa0/5 to see how losing its direct connection to the root bridge would affect the topology.
Verified S3's Fa0/4 became the forwarding root port to confirm the backup path activated.
Repeated the ping test to confirm connectivity recovered after the failure.
Restored S3's Fa0/5 and checked port roles to confirm the preferred path returned.
Saved the switch configurations